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Project Report for Applications of Data Science

Comparative Analysis of Claude 4.5 Haiku, Fine-Tuned SBERT XLM RoBERTa, and Mistral RAG for an Austrian Law Assistant

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1 Choice and pre-training of the LLMs

After careful consideration, the following three models were selected for this project.

1.1 Claude 4.5 Haiku (API)

This model was chosen as a baseline, as an API connection does not efficiently allow for fine-tuning. Although Amazon claims to be the only service provider offering the fine-tuning of Claude models using Amazon Bedrock on custom datasets [6], this option was not utilized. The model is highly powerful, and due to its size, running it locally would not have been possible because of capacity constraints. Claude 4.5 Haiku belongs to Anthropic's class of small, fast models. It is a hybrid reasoning large language model with a knowledge cutoff date of February 2025, meaning its knowledge is most reliable up until that point. It was trained on a proprietary mix of large, diverse datasets, consisting of publicly available internet information up to February 2025, private data from third parties, data from data-labeling services and paid contractors, and data from Claude users who agreed to share their data for training. The final portion of the training data was internally generated by Anthropic. Following the initial pre-training, the model was post-trained and fine-tuned using techniques such as reinforcement learning with both human and AI feedback [1]. These characteristics made it a suitable baseline model.

During inference, the temperature was set to 0.0 to ensure the output remained highly factual and non-creative, thus generating the most probable answer consistently via greedy encoding. The output was additionally constrained to a maximum of 64 tokens for tasks requiring only citation extraction, ensuring the model merely lists the correct citations. For the alternative mode, the maximum tokens were set to 300 to accommodate full textual answer generation, including the citation paragraphs.

1.2 Mistral 7B Instruct v0.1 (4-bit quantized)

Mistral 7B was selected as the second model for multiple reasons. First, it can be run locally by deploying Unsloth in Python to quantize the model from 16-bit to 4-bit. This reduction makes the model manageable to run on an NVIDIA RX 3070 or 3080 GPU. Second, as a French product, the Europe-based origin of the LLM was appealing. Third, the model is open source,

meaning the entire architecture is available for customization and integration under the Apache 2.0 license. Mistral 7B is a Mistral computer based intelligence model that has performed exceptionally well across multiple language projects. It offers high performance while remaining highly resource-efficient. The model underwent extensive training on a large dataset containing more than 1.56 trillion tokens and features 7 billion parameters [5].

During inference, 4-bit quantization was applied via the bitsandbytes and Unsloth libraries to fit the model within the 10GB VRAM of the GPU. For the sampling approach, the default greedy encoding was utilized, and the temperature parameter remained unaltered. The max_new_tokens parameter was set to 64 for strict paragraph citation generation and 250 tokens for the RAG agent. A lower token limit proved insufficient to generate full textual answers, while a significantly higher limit provided too much space for hallucination, potentially leading to non-existent full textual citations given that the model was only provided with paragraph citations.

1.3 SBERT Legal XLM RoBERTa Base Hugging Face Model

The base of this model is XLM-RoBERTa, which is a transformer-based multilingual masked language model. It was pre-trained on text in 100 different languages, distinguishing it from the majority of transformers that were trained exclusively on English text. This multilingual capability makes it an optimal candidate for assessing Austrian legal texts. This model was trained on 2.5 TB of data generated via CommonCrawl across 100 different languages. Conneau et al. [2] demonstrated that XLM-RoBERTa provides substantial gains compared to previous multilingual models, such as mBERT and XLM, across tasks including classification, sequence labeling, and question answering. The vocabulary size is 250K. The XLM-R Base utilized in this work features (L = 12, H = 768, A = 12, 270M params).

The architecture also incorporates Sentence-BERT (SBERT), a modification of the standard BERT model. SBERT focuses on large-scale semantic similarity comparison, clustering, and information retrieval. It is highly applicable to tasks where standard BERT models are no longer computationally feasible. This model reduced the computational time required to find the most similar pair within a collection of 10,000 sentences from 65 hours (using BERT or RoBERTa) to merely 5 seconds, while maintaining the same level of accuracy. SBERT was trained on a combination of the SNLI and Multi-Genre NLI datasets. The SNLI consists of 570,000 sentence pairs, each annotated with the labels contradiction, entailment, or neutral. The MultiNLI dataset

contains 430,000 sentence pairs spanning a wide range of genres. The underlying BERT model was initially trained on Wikipedia. SBERT was trained using a batch-size of 16, a learning rate of 2e-5, a linear learning rate warm-up over 10% of the training data, and the Adam optimizer. It features 12 layers [4].

A Hugging Face ensemble model, authored by Stern5497, combined the XLM-RoBERTa model with SBERT and pre-trained it on a legal corpus. This model is available on Hugging Face under "Stern5497/sbert-legal-xlm-roberta-base". This unique combination creates a sentence-transformer model that maps sentences and paragraphs into a 768-dimensional vector space. It is highly suitable for clustering and semantic search tasks. Since there was no explicit mention of the specific legal dataset used for this pre-training, the model was further fine-tuned on Austrian legal cases to ensure it contained the necessary domain-specific information.

2 Methodology

2.1 Fine-Tuning

To improve accuracy, one of the chosen models was fine-tuned on Austrian tax law cases. The previously mentioned SBERT Hugging Face model was selected for this purpose. Because its foundational knowledge and name (SBERT Legal XLM RoBERTa Base Hugging Face Model) suggest prior training on legal data, it was anticipated that further fine-tuning on Austrian tax law cases would improve its performance. Data was gathered using an API from data.bka.gv.at, the Austrian central open government database, which contains an extensive collection of legal cases from Austrian courts. All available legal cases from the VwGH related to the Einkommensteuergesetz (EStG), Allgemeines Sozialversicherungsgesetz (ASVG), Bundesabgabenordnung (BAO), Doppelbesteuerungsabkommen (DBA), Grunderwerbsteuergesetz (GrEStG), Körperschaftsteuergesetz (KStG), Umsatzsteuergesetz (UStG) were collected.

Legal cases from all these areas were included because utilizing larger and more diverse datasets for fine-tuning generally improves the model's generalization capabilities. The underlying legal cases were extracted alongside their respective paragraph citations, formatted as assigned by the court (e.g., §8 Abs. 2 KStG 1988; §10 Abs. 1 Z 1 KStG 1988). These citations were utilized as ground-truth labels for fine-tuning. Overall, more than 7.500 rows of data spanning the time period 2000-2026 were collected. Figure 1 illustrates an example of the dataset collected via the API. The input consistently consists

of the legal case facts, while the output contains the specific citations used to resolve the matter.

shape: (1_587, 4)

id	instruction	input	output
str	str	str	str
"JWR_2019150007_20210610J01"	"Analysiere den folgenden steue...	"1 Die revisionswerbende Gesell...	"EStG 1988 §4 Abs1; EStG 1988 §...
"JWR_2004150001_20090422X01"	"Analysiere den folgenden steue...	"Im August 2000 langte beim Fin...	"; EStG 1988 §47 Abs2"
"JWR_2008150335_20100520X01"	"Analysiere den folgenden steue...	"Mit dem im Instanzenzug ergang...	"; EStG 1988 §16; EStG 1988 §16...
"JWR_2006150238_20100520X01"	"Analysiere den folgenden steue...	"Beim Beschwerdeführer, einem E...	"; EStG 1988 §4"
"JWR_2017150066_20181219L01"	"Analysiere den folgenden steue...	"1 Der ursprünglich in den USA ...	"; EStG 1988 §34 Abs8"
...	...	...	...
"JWR_2009150194_20120426X01"	"Analysiere den folgenden steue...	"Der Beschwerdeführer erwarb 19...	"; EStG 1988 §2 Abs3 Z6; EStG 1...

Figure 1: Example of dataset collected from data.bka.gv.at using API.

To align with the dataset requirements of the course, only the first 2.000 characters were extracted from each legal case, as some documents exceeded 10 pages in length. Furthermore, unique IDs were used to join multiple instances of legal cases that addressed the same underlying issue but appeared in court multiple times. A custom method was developed to consolidate all paragraph citations that surfaced across these multiple court rounds.

For the fine-tuning process, max_seg_length was set to 512 to prevent VRAM memory exhaustion, which had been an issue in prior iterations. The batch size was set to 4 to prevent GPU instability, and the number of epochs was limited to 1, given the relatively small dataset of just under 10.000 cases. Warm-up steps were set to 100. Within the DataLoader, which prepares the data on the CPU before transferring it to the GPU, shuffling was enabled to ensure the model received a randomized distribution of data sources.

Multiple Negative Ranking Loss was employed for fine-tuning. This function shapes the embedding space by positioning legal cases in close proximity to their corresponding paragraph citations, thereby improving the model's ability to retrieve accurate results for unseen cases. The paragraph citations were unrolled; if a single legal case possessed multiple citations, each citation was assigned as a separate pair with the case text. Using a batch size of 4 ensures that each batch contains one positive match (the legal case and its ground truth) alongside 3 hard negatives, optimally structuring the embedding space. Following fine-tuning, the resulting model was saved so its updated weights and biases could be utilized for subsequent applications.

2.2 RAG-based model

A Retrieval-Augmented Generation (RAG) architecture was implemented by combining the baseline Mistral 7B model for response generation with the fine-tuned RoBERTa model (short name for SBERT Legal XLM RoBERTa Base) for retrieval. RoBERTa serves as an effective retrieval model because it was previously fine-tuned specifically on the Austrian legal case corpus. Paragraph citations were utilized for indexing. While the official Gesetzbücher (law books) from the Austrian legislature were considered, they were not available in a computationally feasible format. Consequently, the approach relied on indexing paragraph citations, effectively configuring the response generation model to act as a legal assistant that directs the user to specific citations and sources. Because Mistral had not been pre-trained on Austrian legal data, it was critical to prevent hallucinations; therefore, strict system instructions were provided to prohibit the fabrication of legal text. Chunking of the indexed documents was unnecessary, as the citations were inherently short.

To enable the RoBERTa model to retrieve accurate results, a FAISS (Facebook AI Similarity Search) index was constructed for the dataset. FAISS is a library utilized for large-scale vector similarity search that scales to millions of vectors, enabling efficient nearest-neighbor retrieval in high-dimensional embedding spaces and generally outperforming traditional database search methods [3]. All paragraph citations were indexed using FAISS. A list of all unique paragraph citations was extracted from the dataset, and these distinct, short citation strings functioned as the document database. Rather than indexing raw citations immediately, embeddings were normalized, and because the text strings were naturally concise, text chunking was bypassed. The citations were then converted into encoded values and loaded into the FAISS index using a batch size of 32.

When a legal case is submitted to the system, it encodes the query and returns the top-k most similar results, alongside a similarity score for each of the k items. These top-k most similar results (in most cases, 5 results with a maximum token size of 300) are subsequently passed from the retrieval model to the response generation model.

3 Results and Evaluation

A total of 4 different evaluation metrics were utilized, as each focuses on a distinct aspect of text prediction.

Citation recall focuses strictly on how often the exact correct citation is

returned in the expected format.

The second metric employed was the BLEU Score, which focuses on exact vocabulary matching. This metric evaluates how many of the n-grams included in the generated prediction appear in the ground truth answer. Its primary advantage is computational speed; however, its main disadvantage is a lack of semantic understanding. Consequently, the BLEU Score performs poorly when correct answers are paraphrased.

The third metric was the ROUGE-L F1 score, which focuses on maintaining the same logical word structure. ROUGE-L identifies the longest common subsequence of words that appears in both the prediction and the correct answer. While it can skip intermediate words and continue searching for the longest word sequence in the correct order, it similarly lacks semantic understanding and struggles with paraphrasing, though it remains computationally efficient.

The final metric utilized was BERTScore. This metric captures semantic equivalence, specifically focusing on identical legal meaning. It processes both the correct answer and the predicted answer through a BERT-based neural network, converting the text into high-dimensional vectors. By comparing the distances between these vectors, BERTScore better reflects the underlying semantic meaning. Although it is computationally more expensive, this limitation is effectively mitigated when utilizing GPUs.

The F1 score, representing the harmonic mean of precision and recall, was selected as the most appropriate option to accurately evaluate the generated answers. Since both high precision and high recall are desired, the F1 score serves as the optimal metric.

Model Setup	**Citation Recall**	**BLEU Score**	**ROUGE-L (F1)**	**BERTScore-F1**
Claude (API)	2.15%	2.40%	16.99%	72.14%
RoBERTa Fn-Tn	0.00%	0.38%	4.52%	59.47%
RAG	3.20%	1.74%	14.00%	68.75%

Table 1: Performance comparison of models across retrieval and generative metrics.

As demonstrated in Table 1, the highest citation recall was achieved by the RAG model, which consists of the fine-tuned legal SBERT XLM-RoBERTa combined with Mistral 7B. It outperformed Claude 4.5 Haiku, despite Claude having more parameters and being trained on a significantly larger dataset. This indicates that domain-specific fine-tuning can improve performance, although the absolute differences in results remain relatively marginal.

The highest BLEU score was attained by Claude 4.5 Haiku. Similarly, the highest ROUGE-L F1 score and the highest BERTScore-F1 were both

achieved by Claude 4.5 Haiku. Consequently, Claude 4.5 Haiku emerged as the overall best-performing model across multiple metrics. Nevertheless, the performance gap between Claude 4.5 Haiku and the Mistral-RoBERTa RAG agent was relatively small, which is an encouraging outcome.

Conversely, the fine-tuned RoBERTa model alone did not perform as well. This outcome primarily stems from the fact that its FAISS index only accesses the paragraph citations rather than the underlying legal text. Essentially, the model was only provided with a formal phrase that included the returned paragraph citations, lacking any contextual explanation of the text. As a result, it could not achieve high scores in vocabulary-focused or semantic metrics. The only metric where it theoretically could have performed better was citation recall; however, it failed to do so in practice.

3.1 Error Analysis

A primary limitation identified during evaluation was that the citation recall metric does not accommodate partially correct answers. For instance, if the model correctly predicts §7 KStG but omits Abs 1, or conversely, if the correct answer is §7 KStG but the model predicts §7 Abs 1 KStG, the prediction is still marked as incorrect. Furthermore, minor formatting discrepancies—such as predicting "Abs" instead of "Abs."—are penalized as incorrect, which misrepresents the model's actual accuracy. This specific formatting issue was particularly prevalent in outputs generated by Claude. As illustrated in Figures 2 and 3, the generated output paragraph citations are nearly identical to the ground truth, differing only by a single punctuation mark.

```
--- Claude 4.5 Haiku | ERROR CASE 1 | ID: CORP-TAX-001 ---
PROMPT (Legal Case) : Was ist die steuerliche Bemessungsgrundlage für die Körperschaftsteuer?
REQUIRED CITATION   : § 7 Abs. 1 KStG 1988
CORRECT ANSWER      : Das Einkommen, das der unbeschränkt Steuerpflichtige innerhalb eines Kalenderjahres bezogen hat (§ 7 Abs. 1 KStG).
MODEL GENERATED     : Die Bemessungsgrundlage der Körperschaftsteuer ist das Einkommen der Körperschaft, das nach den Vorschriften des E
SCORES              : BERTScore: 68.40% | ROUGE-L: 15.73% | BLEU: 0.72%
```

Figure 2: Example of poor text notation versatility robustness - correct answer.

```
ıpitalvermögen, Vermietung und Verpachtung sowie sonstigen Einkünften zusammensetzt (§ 7 Abs 1 KStG 1988). Von diesem Einkommen werden Bet
```

Figure 3: Example of poor text notation versatility robustness - predicted solution.

Another frequent issue, primarily associated with Claude, occurred when the model correctly identified the applicable law book (e.g., KStG) but cited the incorrect paragraph. Despite this, the prediction received a relatively high BERTScore, indicating that the model generated a semantically accurate description of the solution while failing at exact citation retrieval.

A notable problem with the fine-tuned RoBERTa model was its occasional tendency to hallucinate or generate entirely inaccurate solutions. For instance, as demonstrated in Figure 4, the model incorrectly suggested the tax law EStG Preussen 1891 §8, dating back to the year 1891. In another

```
--- FineTuned RoBERTa | ERROR CASE 6 | ID: CORP-TAX-006 ---
PROMPT (Legal Case) : Wie hoch ist der maximal mögliche Verlustabzug in einem Veranlagungsjahr und welche Ausnahmen gibt es?
REQUIRED CITATION   : § 8 Abs. 4 Z 2 lit. a–b KStG 1988
CORRECT ANSWER      : 75 % des Gesamtbetrages der Einkünfte (§ 8 Abs. 4 Z 2 lit. a KStG). Ausnahmen (volle Verrechnung): u.a. Sanierungs
MODEL GENERATED     : Nach sorgfältiger Prüfung Ihres Falles verweise ich auf die folgenden Paragraphen: EStG Preußen 1891 §8; Durchschn
SCORES              : BERTScore: 62.15% | ROUGE-L: 5.48% | BLEU: 0.00%
```

Figure 4: Example of hallucinated solution.

instance, the fine-tuned RoBERTa model advised the application of DBAbk USA, DBAbk Argentinien, and several other treaties concerning the Doppelbesteuerungsabkommen (Double Taxation Agreements), ultimately failing to answer the specific query.

These retrieval errors consequently impacted the predictions of the broader RAG architecture. When the fine-tuned RoBERTa model misunderstood the semantic meaning of a query and retrieved irrelevant citations, the Mistral generation model was forced to synthesize an answer based on fundamentally flawed context. As shown in Figure 5, a query involved the term 'Mantelkauf', a legal concept which possesses a specific meaning distinct from its literal translation of buying a coat. Because RoBERTa failed to comprehend this domain-specific meaning, it could not retrieve the correct citations, thereby negatively influencing the text generation phase.

```
--- RoBERTa + Mistral RAG | ERROR CASE 7 | ID: CORP-TAX-007 ---
PROMPT (Legal Case) : Was ist ein Mantelkauf und welche steuerliche Konsequenz hat er?
REQUIRED CITATION   : § 8 Abs. 4 Z 2 lit. c KStG 1988
CORRECT ANSWER      : Ein Mantelkauf liegt vor, wenn die wirtschaftliche Identität einer Körperschaft durch wesentliche Änderung der Ges
MODEL GENERATED     : Ein Mantelkauf ist eine Spende, die in Form eines Gegenwerts erfolgt.
Die Spende ist steuerlich absetzbar, wenn sie an einen gemeinnützigen Zweck geht (§ 1 Abs 1 BewG 1955).
Das Gegenwert wird als Verbrauchsausgabe gewertet und ist steuerlich absetzbar, wenn es nicht mehr als 10% des Gesamtwertes des Mantels

### Zitierungen:

- § 1 Abs 1 BewG 1955
- § 1 Abs 2 BewG 1955
SCORES              : BERTScore: 68.39% | ROUGE-L: 12.50% | BLEU: 1.05%
```

Figure 5: Example of retrieval model negatively influencing text generation Mistral model.

4 Conclusion

In conclusion, this work demonstrated effective methods for collecting domain-specific data from various sources, particularly through the utilization of APIs. Furthermore, it provided practical insights into utilizing NVIDIA GPUs for the local execution and fine-tuning of Large Language Models (LLMs).

The performance of the fine-tuned model was somewhat underwhelming compared to initial expectations. Achieving a 0% in citation recall and correspondingly low scores across all other metrics was unexpected, given the volume of legal cases processed during training. However, from the perspective of deep neural network training, a dataset comprising only a few thousand cases remains relatively small. Future methodologies should consider collecting a substantially larger dataset and employing sampling techniques, such as oversampling or undersampling, to create a more balanced distribution. This would prevent the model from overfitting to a single law book, as the number of available cases and the sources of paragraph citations varied significantly across the dataset. Additionally, scraping the relevant law books and integrating their full texts into the FAISS index for the fine-tuned RoBERTa model could potentially improve retrieval scores.

Furthermore, modifying the citation recall metric to enhance its robustness against minor formatting variations in the generated answers would yield a more accurate evaluation.

The deployment of different, larger LLMs—specifically those trained on more extensive datasets and possessing a greater number of hyperparameters—could also yield superior results. Implementing this, however, would require further investigation into the feasibility and technical implementation of distributing the computational load for model loading and fine-tuning across multiple NVIDIA GPUs with differing VRAM capacities.

There remains significant scope for future exploration in this domain, particularly regarding the testing of alternative models, diverse hyperparameter configurations, and varied datasets. Ultimately, this project demonstrated that larger models generally produce better results, as evidenced by Claude 4.5 Haiku consistently achieving the highest scores. Nevertheless, the custom-built RAG agent performed competitively when compared to this significantly more advanced LLM.

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